Word => sōz, biz ōqiydigan bōlak, bōsh joy bilan ajratiladi.

Token => modelning ōqiydigan bōlagi. U sōz, sōzning qismi, yoki bitta belgi bōlishi m-n

English text => 1 token ≈ 0.75 words.

Case 1: Token == Words.

Text => "Cats sit on a mat"

words: cat, sit, on, a, mat => 5ta

tokens: cat, sit, on, a, mat => 5ta

Case 2: Token > Words

=> Uzun yoki kam uchraydigan sōz bōlaklarga bōlinadi.

=> Tinish belgilari alohida token bōladi

=> ōzbek, koreys kabi tillar kōproq tokenga b-di.

Text: "Deconstitutionalitation".

words => 1ta

tokens => De, consti, tutional, iza, tion. 5ta

Case 3: Token < Words.

Text: "United States"

words: United, States => 2ta

tokens: United States => 1ta.

- bu sōzni bir biridan ajratsa ma'no qolmaydi.

lug'atda bu bitta butun sōz.

=> English - 1 token ≈ 0.75 words.
=> Spanish/French - 1 token ≈ 0.50 words.
=> Russian - 1 token ≈ 0.28 words.
=> Japanese/Korean - 1 token ≈ 0.18 words.
Ingliz tili uchun token/so'z kam,
Chunki tokenizer shu tilde ōgitilgan.

LLM model training qilishdan oldin hisoblash
kerak bōlganlar.

① Parameters: $\boxed{12 \times L \times d^2}$

- Layers => L -> transformer layerlar soni.
- d => hidden dimension, ya'ni bitta tokenni
  ifodalovchi vektor o'lchami.
- d=768 => bitta token model ichida 768 ta son
  qiymatdan iborat vektor bilan ifoda
- bitta layer uchun taxminan $12d^2$ para
  chiqadi.

Misollar: "Hello world" => ["Hello", "world"

Agar d=4. bōlsa,
Model "Hello" tokenini quyidagicha ifod

Token: Hello 0.7 -0.1 0.5
0.2 4ta son -> d=4, 1ta toke
d=768 bōlsa

Agar:
tion + FFN makit

Nega d^2?

Model shu vektorni ōzgartirish uchun (weight matrix) ishlatadi. $d=4$ bōlsa matrix ōlchami $4 \times 4 = 16$ ta weight bōlishi mumkin. $\underline{d^2 = 16}$

Jami parameters

$L = 24$ $\qquad 12 \times d^2 \times L$

$d = 2048$ $\qquad 12 \times 2048^2 \times 24 \approx \boxed{1.21 \text{ milliard}}$

② Xotira sig'imi:

$$\boxed{16\ B \times N}$$

$$\left.\begin{array}{l} \text{Weights (FP32) - 4 bayt} \\ \text{Gradients (FP32) - 4 bayt} \\ \text{Optimizer states - 4 bayt} \\ \text{Activations - 4 bayt} \end{array}\right\} \rightarrow \underline{16\ \text{Bayt}}$$

$$\boxed{\text{Memory} = N \times \text{har parametr uchun bayt}}$$

Memory = 16×1.21 milliard ≈ 19.3 GB.

Agar bizda 320 GB mavjud bōlsa, taxminan ⑥%

③ Compute budget.

$$\boxed{C = 6 \cdot N \cdot D}$$

- N - model parameters soni.
- D - training tokenlar soni.

Chinchilla qoidasi bōyicha $\boxed{D \approx 20 \times N}$

- Forward = 2, Backward ≈ 4. → $\textcircled{6ND}$.

$C = 6 \times 1.21$ milliard $\times$ 24.2 milliard

$\underline{1.75 \times 10^{20}}$ FLOPs

④ Vaqt (wall-clock)

$$\boxed{t = \frac{6ND}{GPU \cdot peak \cdot MFU}}$$

- peak - A100 BF16 = 312 TFLOPs/s
- MFU - real samaradorlik, A100 da 40%.
- GPU - 4 ta parallel ishlay.

$t = 1.75e20 / (4 \cdot 0.40 \cdot 312e12) \approx 4$ kun.

4 kun ($\approx$ 97 soat).

⑤ Cost:

$$\boxed{Cost = GPU_{soni} \times Soat \times GPU_{narx/soat}}$$

GPU soni - 4 ta.

GPU narxi - $1.5-2 / soat

Cost = 97 × 4 × 1.5-2 ≈ $600 - 800

Bitta to'liq o'qitish siklining narxi.

Trade off => biron narsaga erishish uchun biron narsadan voz kechish.

Parametrlar	$N \approx 12 \times L \times d^2$
Xotira (AdamW)	16 bayt × N
Compute	$C = 6 \times N \times D$
tokenlar (chinchilla)	$D \approx 20 \times N$
Vaqt	$6ND / (GPU \cdot peak \cdot MFU)$
Narx	Soat × GPU × $/soat

Params => Xotira => Compute => Vaqt => Narx => Samarador

Trainingdan oldin.

Sonlarni kompyuterda ifodalash uchun ishlatiladigan (data formatlar). Ular sonning qancha xotira egallashi va hisoblash aniqligiga ta'sir qiladi.

$\Downarrow$

FP32 => 32 bit Floating point.
FP -> Floating point, ya'ni kasrli sonlarni ifodalar.
Xotirada har bir son 4 byte.
1e-38 ... 1e38.

FP16 => 16bit => 2 byte -> 6e-5 .. 65504.
BF16 => 16bit => 2byte -> FP32 b-n bir xil

Bitlar taqsimoti:

FP32 -> 32 Bit
Sign -> 1 bit, Exponent -> 8 bit, Fraction - 23bit.
FP16 -> 16 bit.
Sign -> 1 bit, Exponent -> 5 bit, Fraction - 10bit.
BF16 -> 16 bit
Sign 1bit, Exponent -> 8 bit, Fraction - 7 bit.
Sign -> son musbat yoki manfiyligini aniqlaydi.
Exponent -> sonning kattalik diapazonini aniqlaydi.
Fraction -> Sonning aniqliligini aniqlaydi.

=> Trainingda gradientlar juda kichik bo'lishi mumkin. (10^{-8}). FP16 da bu aniq nolga aylanadi. Chunki ~~Fraction~~. diapazoni kichik

$6.1 \times 10^{-5} > 10^{-8}$ => gradient -> 0

BF16/FP32 da to'g'ri saqlanadi. diapazon katta

Xulosa: BF16 => tezlik, FP32 => xavfsizlik
FP16 -> xavfli.

Param

Model weights (FP32) -> 4 byte
Gradients (FP32) -> 4 byte.
Adam m (FP32) -> 4 byte.
Adam v (FP32) -> 4 byte.
Jami (AdamW) -> 16 byte